SPECTRAL SPACES ARE EXACTLY INVERSE LIMITS OF FINITE SOBER SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the inverse limit of a directed system of finite sober spaces is spectral. Invoked at the start of the counted proof (source lines 4927-4928). Gives one direction.
- The description of a spectral space as a subset of a product of two-point spaces closed in the constructible topology. Invoked inside the counted proof (source lines 4929-4933). Realises the space as a limit over the finite subsets of the index set.

Source reference. Original lemma lemma-directed-inverse-limit-finite-sober-spectral-spaces: sources/stacks-topology/pinned-original-source.tex, source lines 4851–4851. Navigation pointer only: the counted text refers to the Stacks item lemma-directed-inverse-limit-finite-sober-spectral-spaces, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-spectral-closed-in-product-two-point-space: sources/stacks-topology/pinned-original-source.tex, source lines 4877–4877. Navigation pointer only: the counted text refers to the Stacks item lemma-spectral-closed-in-product-two-point-space, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *A topological space is spectral if and only if it is a directed inverse limit of finite sober topological spaces.*

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Proof. One direction is given by Lemma lemma-directed-inverse-limit-finite-sober-spectral-spaces. For the converse, assume X is spectral. Then we may assume $X \subset \prod_{i \in I} W$ is a subset closed in the constructible topology where $W = \{0, 1\}$ as in Lemma lemma-spectral-closed-in-product-two-point-space. We can write

$$\prod\nolimits_{i \in I} W = \lim\nolimits_{J \subset I \text{ finite}} \prod\nolimits_{j \in J} W$$

as a cofiltered limit. For each J, let $X_J \subset \prod_{j \in J} W$ be the image of X. Then we see that $X = \lim X_J$ as sets because X is closed in the product with the constructible topology (detail omitted). A formal argument (omitted) on limits shows that $X = \lim X_J$ as topological spaces. $\square$